

Chapter 3:

Orthonormal Basis, Bra–Ket Notation, and Measurement

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Orthonormal Basis

- Each basis has its set of basis vectors, e.g. $\hat{x}$ and $\hat{y}$ **span** a plane and they form the basis of a 2D plane.
- Any vector on the 2D plane can be represented by a **linear combination** of the basis vector, $\hat{x}$ and $\hat{y}$.
- Vectors are equivalent to the states of a quantum system. So, states are vectors and basis states are basis vectors.
- If we add $\hat{z}$, then $\hat{x}$, $\hat{y}$ and $\hat{z}$ are the basis vectors and they span the 3D space.
- Any 3D vector can be represented by a linear combination of $\hat{x}$, $\hat{y}$ and $\hat{z}$.
- The basis vectors can be labeled as $\widehat{v}_0$ to $\widehat{v}_{n-1}$ and any vector in this space can be represented as a linear combination of $\widehat{v}_0$ to $\widehat{v}_{n-1}$.
- An electron's spin state (vector) must be a linear combination of some basis states (vectors). One of the possible sets of a basis vectors are $|spin\ up\rangle$ and $|spin\ down\rangle$, which its dimension of the spin space is 2D because it has 2 basis vectors.
- However, in a real 2D space, the coefficients in the linear combinations are all real numbers. In spin space, the coefficient can be complex numbers.

Orthonormal Basis

- One type of bases is of particular importance, called the **orthonormal basis**.
- An orthonormal basis is defined as:

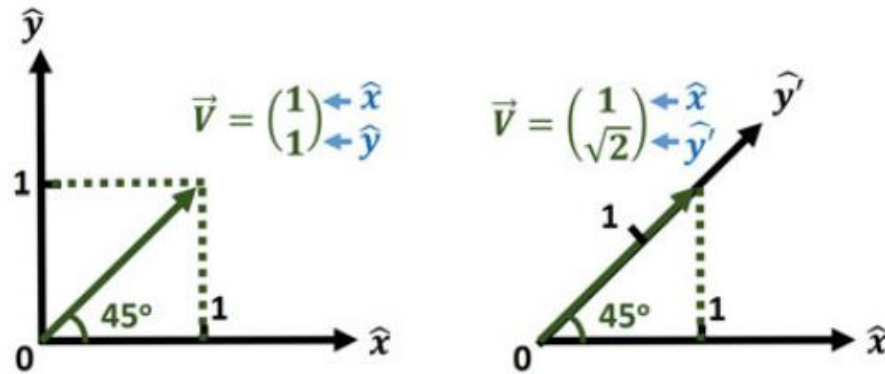
$$\widehat{v}_i \cdot \widehat{v}_j = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} = \delta_{ij}$$

where $\widehat{v}_i$ and $\widehat{v}_j$ are any two basis vectors, i and j can be any number from 0 to $n - 1$, if it is an n -D space.

- E.g. if it is our regular 3D space, we may set $\widehat{v}_0 = x$, $\widehat{v}_1 = y$, $\widehat{v}_2 = z$. So, $\widehat{v}_i$ and $\widehat{v}_j$ can be x and y , or x and z , or y and z , or y and x , or z and x , or z and y , and can also be x and x , or y and y , or z and z .
- From the above equation, says that if the inner product of every basis vector with itself is one (it is **normalized**, with $|L| = 1$) and the inner product with other basis vector is zero, then this set of basis vectors and the basis formed by this set of basis vectors are orthonormal (*orthogonal and normalized*).
- If the basis vectors are orthonormal, it means all of them have no common part.
- δ_{ij} is called the **Kronecker delta**. It is just a notation. It means that if $i = j$, it is one. If $i \neq j$, it is zero.

Orthonormal Basis

- Two different sets of basis vectors, orthonormal (left) and non-orthonormal (right).



- $\vec{v}$ can be represented as $|\vec{v}\rangle = 1|\hat{x}\rangle + 1|\hat{y}\rangle = 1|\hat{x}\rangle + \sqrt{2}|\hat{y}'\rangle$
- The $\hat{x}$ and $\hat{y}'$ basis is very difficult to deal with. In quantum computing, we will always use an orthonormal basis.
- We can check if $\hat{x}$ and $\hat{y}$ are orthogonal by observing their inner product.
- Since $\hat{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\hat{y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, their inner product is:

$$\hat{x} \cdot \hat{y} = (1 \ 0) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 \times 0 + 0 \times 1 = 0$$

Bra-Ket Notation

- For every vector $\vec{v}_a$ (or this is just the quantum state), we can represent it as $\vec{v}_a$, “Vector a”, $|\vec{v}_a\rangle$, $|v_a\rangle$, $|a\rangle$, $|vector\ a\rangle$, etc.
- If we write it in the form of $|\ \rangle$, then it is Dirac’s bra-ket notation. $|\ \rangle$ is just a vector and we can perform any vector operations on it.
- For example, for $\vec{v}_a = \vec{v}_b + \vec{v}_c$, we can write as $|\vec{v}_a\rangle = |\vec{v}_b\rangle + |\vec{v}_c\rangle$ or $|a\rangle = |b\rangle + |c\rangle$.
- How about the **bra**? It still represents a vector and it has the form of $\langle \ |$.
- We can represent $\vec{v}_\alpha = \vec{v}_\beta + \vec{v}_\gamma$ in the form as $\langle \vec{v}_\alpha | = \langle \vec{v}_\beta | + \langle \vec{v}_\gamma |$.
- The set of vectors in the *bra-representation* is the same set of vectors in the *ket-representation* and they form a **dual space**. Each vector in the ket space can find a corresponding vector in the bra space.
- The method to find the correspondence is by performing a transpose operation and then complex conjugation.
- E.g.

$$|v_1\rangle = |1\rangle = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 + 2i \\ i \end{pmatrix} \rightarrow \langle v_1| = \langle 1| = (a^* \ b^*) = (1 - 2i \quad -i)$$
- Therefore, $\vec{v}_1 \cdot \vec{v}_2 = \langle v_1 || v_2 \rangle = (a^* \ b^*) \begin{pmatrix} c \\ d \end{pmatrix} = a^* \times c + b^* \times d$

Quantum Mechanical State, Superposition, and Measurement

- Every vector is a state of a physical system. If we have a physical system called electron spin and it has 2 basis vectors, which are $|spin\ up\rangle$ and $|spin\ down\rangle$. We can call them $|\uparrow\rangle$ and $|\downarrow\rangle$ also.
- The basis vectors themselves are vectors and also quantum mechanical states. But they have a special identity called basis vectors because we choose them to represent other vectors as linear combinations of them.
- We can write the basis state as follows:

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad and \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
- Any spin state of an electron can be written as $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$.
- Any electron spin state is a superposition, which we have been calling “linear combination” of $|\uparrow\rangle$ and $|\downarrow\rangle$.
- In quantum mechanics, the spin state $|\psi\rangle$ is a linear combination of $|\uparrow\rangle$ and $|\downarrow\rangle$. But we know that $|\uparrow\rangle$ and $|\downarrow\rangle$ must be orthogonal and cannot exist at the same time with 100% certainty.
- As a superposition state, when we perform a measurement, the spin state $|\psi\rangle$ will collapse to one of the basis states. The probability of collapsing to $|\uparrow\rangle$ or $|\downarrow\rangle$ is *proportional* to $|\alpha|^2$ and $|\beta|^2$, respectively.

Quantum Mechanical State, Superposition, and Measurement

- The important meaning of the coefficients in the superposition is due to the **Born rule** which is an essential part of the **Copenhagen interpretation**.
- For a state $|\psi\rangle$ in an n -D space, which can be represented as

$$|\psi\rangle = a_0|V_0\rangle + a_1|V_1\rangle + \cdots + a_{n-1}|V_{n-1}\rangle = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix}$$

where $|V_i\rangle$ are the basis states, when we perform a measurement, $|\psi\rangle$ will randomly collapse to one of the basis states, e.g. $|V_j\rangle$ with a probability proportional to $|a_j|^2$.

- It equals $a_j^* a_j$, where a_j^* is the complex conjugate of a_j .
- For example, if $a_j = 1 + i$, then $a_j^* = 1 - i$ and $a_j^* a_j = (1 + i)(1 - i) = 1 \times 1 + i \times 1 + 1 \times -i + i \times -i = 2$.
- The length of a_j is $\sqrt{2}$ and thus $|a_j|^2 = 2$.
- If a vector is **normalized**, which will be used throughout quantum computing, it means it has a unit length.
- The length of a vector is defined by the inner product of itself.

Quantum Mechanical State, Superposition, and Measurement

- For example, the inner product of $|V\rangle$ is $\langle V|V\rangle = (a_0^* \ a_1^* \ \dots \ a_{n-1}^*) \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix}$
 $= a_0 a_0^* + a_1 a_1^* + \dots a_{n-1} a_{n-1}^*$

$$= \sum_{i=0}^{n-1} a_i a_i^* = \sum_{i=0}^{n-1} |a_i|^2 = 1$$

- $\langle V|V\rangle = 1$ because it is normalized. So, the probability that it will collapse to the basis state $|V_i\rangle$ is proportional to $|a_i|^2$.
- In summary, if a vector (or state) is normalized, the probability that it will collapse to the basis state $|V_i\rangle$ equals to $|a_i|^2$.
- If you are given a vector, $|V\rangle$, which is not normalized, we need to normalize it first. Since the definition of the normalized vector is that the summation of the square of the magnitude of coefficient is equal to one, we can just divide the vector by this sum and it will be normalized (with length = 1) as the following:

$$|V'\rangle = \frac{|V\rangle}{\sqrt{\sum_{i=0}^{n-1} |a_i|^2}}$$

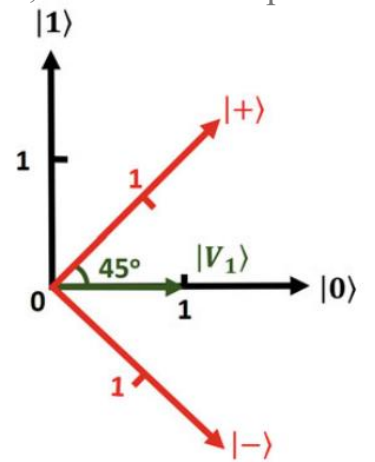
Chapter 4:

Changing Basis, Uncertainty Principle, and Bra-Ket Operations

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Basis and Uncertainty Principle

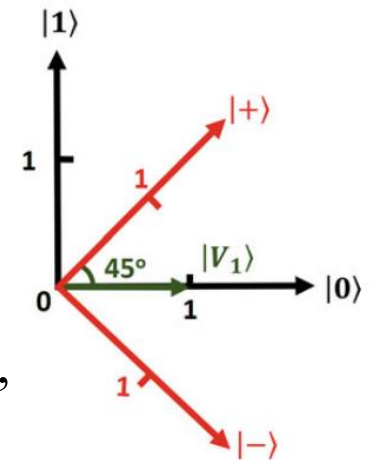
- It is important to realize that $|\uparrow\rangle$ and $|\downarrow\rangle$ are not real space direction vector like $\hat{z}$ and $-\hat{z}$. They are not direction vectors.
- We can also choose another basis with basis vectors as $|\text{spin left}\rangle$ and $|\text{spin right}\rangle$. We write them as $|\leftarrow\rangle$ and $|\rightarrow\rangle$. These vectors have nothing to do with $\hat{x}$ and $-\hat{x}$. $|\leftarrow\rangle$ and $|\rightarrow\rangle$ are *not* real space direction vectors.
- We will also rename the basis vectors. We set $|\uparrow\rangle = |0\rangle$, $|\downarrow\rangle = |1\rangle$, $|\leftarrow\rangle = |+\rangle$, and $|\rightarrow\rangle = |-\rangle$.
- Therefore, the $|\uparrow\rangle/|\downarrow\rangle$ basis becomes the $|0\rangle/|1\rangle$ basis and the $|\leftarrow\rangle/|\rightarrow\rangle$ basis becomes the $|+\rangle/|-\rangle$ basis.
- The vector in the spin space could have complex coefficients when they are represented as the linear combination of $|0\rangle$ and $|1\rangle$, i.e. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β may be complex numbers.
- Let us assume α and β are real, and then we can treat them as vectors on a 2D plane.
- On the 2D plane, we can also have another set of basis vectors which are just equivalent to rotating the $|0\rangle/|1\rangle$ basis vectors clockwise by 45° . We call them $|+\rangle/|-\rangle$ basis as shown in Pic.4.
- From Pic.4, by observation, we can represent $|V_1\rangle$ in the $|0\rangle/|1\rangle$ basis, $|V_1\rangle = 1|0\rangle + 0|1\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and can represent $|V_1\rangle$ in the $|+\rangle/|-\rangle$ basis, $|V_1\rangle = 1/\sqrt{2}(|+\rangle + |-\rangle) = 1/\sqrt{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.



Pic.4 The two bases $|0\rangle/|1\rangle$ and $|+\rangle/|-\rangle$ represented in a 2D real space.

Basis and Uncertainty Principle

- From Pic.4, by observation, we can obtain $|0\rangle$ with 100% certainty from measurement $|V_1\rangle$ in the $|0\rangle/|1\rangle$ basis, but it is completely uncertain to us whether it is spinning left or spinning right in the $|+\rangle/|-\rangle$ basis.
- This means that we cannot measure the spin up/down and spin left/right with 100% accuracy simultaneously. This is called the **Heisenberg Uncertainty Principle**.
- In quantum mechanics, the measurement of momentum and position is incompatible. Therefore, we cannot measure the momentum and position of a particle precisely at the same time.
- Note that measurement of the momentum means to collapse the state into a momentum basis vector, which happens to be the momentum in our daily experience, and measurement of position means to collapse the particle state into a position basis vector, which is the position in our daily experience.
- There is a valid reason why we cannot measure the momentum and position with 100% certainty at the same time. The more certain we have on one measurement, the less we have on another measurement.



Pic.4 The two bases $|0\rangle/|1\rangle$ and $|+\rangle/|-\rangle$ represented in a 2D real space.

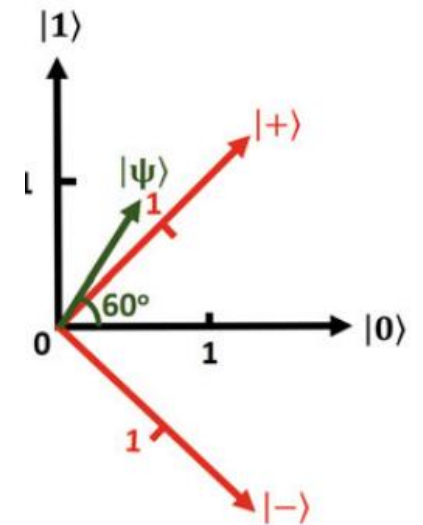
Basis and Uncertainty Principle

- $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$ are represented in the $|0\rangle/|1\rangle$ basis. Its angle to the horizontal axis is 60° .
How do we represent it in the $|+\rangle/|-\rangle$ basis?
- We can use trigonometry, but this is not useful to us when the problem cannot be mapped onto a fictitious 2D plane.
- The trick is to represent $|0\rangle/|1\rangle$ basis vectors as the linear combinations of $|+\rangle/|-\rangle$ basis vectors and then use substitution.
- Firstly, we can represent $|0\rangle$ and $|1\rangle$ as

$$|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \text{ and } |1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$$

- Then we substitute $|0\rangle$ and $|1\rangle$ into $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$ and obtain:

$$\begin{aligned} |\psi\rangle &= \frac{1}{2} \left(\frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \right) + \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \right) \\ &= \frac{1}{\sqrt{2}} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) |+\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \right) |-\rangle \end{aligned}$$



Pic.5 $|\psi\rangle$ in the fictitious 2D real space.

More Bra-ket Operations

- For any vector in the ket space, there is a corresponding vector in the bra space. This is called **Dual Correspondence**.
- For example, if $|\alpha\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix}$, then $\langle\alpha| = (-i \ 1)$
- We can scale a vector $|\alpha\rangle$ by a constant (may be complex number) c . And this is still a vector, $c|\alpha\rangle$; for example, $2\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$.
- What is the corresponding vector in the bra space? It is equivalent to multiplying the complex conjugate of c , i.e. c^* , to the bra version of $|\alpha\rangle$, i.e. $\langle\alpha|$.

$$c^*\langle\alpha| \leftrightarrow c|\alpha\rangle$$

- Similarly, if we represent a vector $|\gamma\rangle$, as a linear combination of other vectors in the ket space, we can find the corresponding vector in the bra space as:

$$\langle\gamma| = a^*\langle\alpha| + b^*\langle\beta| \leftrightarrow |\gamma\rangle = a|\alpha\rangle + b|\beta\rangle$$

More Bra-ket Operations

- There is another very useful rule and it is about the inner product.
- An inner product of two vectors is a scalar (pure number). So, there is no dual space for it.
- However, there is a rule related to the dual space for describing the relationship between $\langle\alpha|\beta\rangle$ and $\langle\beta|\alpha\rangle$.
- $\langle\alpha|\beta\rangle$ means the projection of vector β on vector α and $\langle\beta|\alpha\rangle$ means the projection of vector α on vector β .
- If $|\alpha\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}$ and $|\beta\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$, then $\langle\alpha| = (a_0^* \ a_1^*)$ and $\langle\beta| = (b_0^* \ b_1^*)$, and therefore,

$$\langle\alpha|\beta\rangle = a_0^* b_0 + a_1^* b_1$$

$$\langle\beta|\alpha\rangle = b_0^* a_0 + b_1^* a_1$$

- There is a special case when $\alpha = \beta$. This means $\langle\alpha|\alpha\rangle = \langle\alpha|\alpha\rangle^*$.
- This is just the inner product of vector α to itself and it says that the result equals its complex conjugate. Since only a real number equals its complex conjugate, it means the inner product of any vector to itself must be real.
- If two vectors are orthogonal to each other, their inner product is zero, $\langle\alpha|\beta\rangle = 0$.
- If a vector is normalized, its inner product is 1, $\langle\alpha|\alpha\rangle = 1$.